

Yufan DONG

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EDUCATION

Institute of Neuroscience, CEBSIT, CAS	Shanghai, China
• Ph.D in Neuroscience	September. 2020-2025
Cold Spring Harbor Asia	Suzhou, China
• Summer school of computational and cognitive neuroscience	July. 2023
Central South University	Hunan, China
• B.S. in Biological Science, School of Life Sciences	September. 2016-June. 2020
Hong Kong University	Hong Kong, China
• Summer school of Chemistry	July. 2017-Aug. 2017

PUBLICATIONS

- [Yufan Dong](#), Danqian Liu*. Orchestrating REM sleep with eye movements. 2025, in preparation.
- [Yufan Dong](#), Danqian Liu*. Cracking the complexity of REM sleep. *Trends in Neurosciences*. 2025 Dec; 48(12):977-993.
- [Yufan Dong](#), Danqian Liu*. Awakening from anesthesia switches cortical wave modes. *Trends in Neurosciences*. 2023 July; 46(7):503-504,.
- [Yufan Dong](#), Jiaqi Li, Min Zhou, Yihui Du, Danqian Liu*. Cortical Regulation of two-stage Rapid Eye Movement Sleep. *Nature Neuroscience*. 2022 Nov 18; 25:1675-1682.
- [Yufan Dong](#)‡, Renbin Lu‡, Jing Zhang, Xiushan Wu, Jia-Da Li*. Deficiency in Prader-Willi syndrome gene *necdin* leads to attenuated cardiac contractility. *iScience*. 2024; 27(6):109974.
- Renbin Lu, [Yufan Dong](#), Jia-Da Li*. *Necdin* regulates BMAL1 stability and circadian clock through SGT1-HSP90 chaperone machinery. *Nucleic Acids Res* . 2020 Aug 20; 48(14):7944-7957.
- Xiaoliang Xing‡, Kunyang Wu‡, [Yufan Dong](#), Yimei Zhou, Jing Zhang, Fang Jiang, Wang-Ping Hu, Jia-Da Li*. Hyperactive Akt-mTOR pathway as a therapeutic target for pain hypersensitivity in *Cntnap2*-deficient mice. *Neuropharmacology* . 2020 Mar 15; 165:107816.

CONFERENCE PRESENTATIONS

- 2025 Graduated student symposium of Institute of Neuroscience, CAS (**Oral presentation & Poster**)
Title: REM sleep eye movements ignite global cortical state
- 2024 Asia-Pacific Computational Cognitive Neuroscience (AP-CCN) conference (**Invited talk**)
Title: Untangling the Mystery of REM Sleep: Spontaneous Cortical Dynamics and Their Functions
- 2024 13th Ray Wu Symposium & Ray Wu Prize Award (**Invited talk**)
Title: Untangling the Mystery of REM Sleep: Spontaneous Cortical Dynamics and Their Functions

- 2024 Graduated student symposium of Institute of Neuroscience, CAS (*Oral presentation & Poster*)
Title: *A new paradigm for investigating REM sleep function in problem solving*
- 2023 53rd annual meeting of the Society of Neuroscience (*Poster*)
Title: *Cortical regulation of two-stage REM sleep*
- 2023 The 16th annual meeting of Chinese neuroscience society (*Poster*)
Title: *Cortical regulation of two-stage REM sleep*
- 2021 The annual meeting of Institute of Neuroscience (*Oral presentation*)
Title: *Cortical regulation of two-stage REM sleep*

RESEARCH EXPERIENCES

- **Study of cortical representation of spontaneous orofacial dynamics during REM sleep** (*In preparation*)
ION, State key laboratory of Neuroscience July. 2024-present
Advisor: Dr.Danqian Liu
- **Study of REM sleep function in creative problem solving, one of the higher cognitive functions suggested as highly correlation with REM sleep.** (*Ongoing*)
ION, State key laboratory of Neuroscience Nov. 2022- now
Advisor: Dr.Danqian Liu
- **Study of Cortical dynamics during REM sleep and mechanisms of REM sleep regulation** (*Publication*)
ION, State key laboratory of Neuroscience March. 2021- July. 2022
Advisor: Dr.Danqian Liu
- **Study of Molecular mechanisms of Circadian disorders in PWS (Prader-Willi Syndrome) patients** (*Publication*)
State key laboratory of Medical Genetics June. 2017- May. 2020
Advisor: Prof.Dr.Jiada Li
- **Study of Molecular mechanisms of heart dysfunction in PWS patients** (*Publication*)
State key laboratory of Medical Genetics February. 2018- June. 2020
Advisor: Prof.Dr.Jiada Li
- **TNBC Assassin: genetic therapy circuit targeted TNBC (Triple Negative Breast Cancer)** (*Website*)
Central South University December. 2018- November. 2019
Advisor: Prof.Dr.Jiada Li & Dr.Shanni Li
- **Other research: Targeting Akt-mTOR pathway to rescue the phenotype in Cntnap2-deficient autism mouse model** (*Publication*)
State key laboratory of Medical Genetics July. 2017-June. 2019
Advisor: Prof.Dr.Jiada Li

ACADEMIC HONORS & REWARDS

- RayWu Prize 2024
- CAS President Award 2024
- National Scholarship Oct. 2017/Oct. 2022
- Gold medal in 2019 International Genetically Engineered Machine (iGEM) competition Mar. 2019-Nov. 2019
- First prize in 2021 annual meeting of ION, CAS Dec. 2021
- Excellent Student Scholarship Sept. 2017/Sept. 2018/Sept. 2019
- First & Second Prize of Academic Scholarship Sept. 2017/Sept. 2018

- Second & Third Prize in the 2017 & 2019 National English Competition for College Students *Oct. 2017/Oct. 2019*
- "Top ten excellent poster" in Poster presentation of Collaborative Innovation Center for Genetics and Development Communication *Dec. 2016*
- Winning Prize in the fifth Basic medical experiments and innovative design competition in Central South University *Oct. 2017*
- First Prize in International Symposium on Cryobiology and Biobank & College innovation forum *Oct. 2019*
- 2018 & 2019 National Innovation Project for College Students finished and won "Excellent work" *Mar. 2018-Oct. 2019*

SKILLS

- **Familiar with** MATLAB/python/R/C++ programming and data analysis, Photoshop, AI, Pr, Prisms, regular cognitive neuroscience technique, $\text{\LaTeX}$, etc.
- **Familiar with** Data processing and analysis with existed and original tools from imaging and electrophysiological recording data.
- **Expert at** mice surgery, including open window for large-scale wide-field imaging, two-photon imaging, mini-scope, fiber photometry, optogenetics, etc.
- **Expert at** mice behavioral training, including stereotyped behavioral paradigm and DIY behavioral paradigm.

RESEARCH INTERESTS

- Function of dreaming and rapid eye movement sleep.
- Hierarchy and integration of time and space representation.
- In search of consciousness generation and consciousness level altering mechanisms through inter-layer communication across cortex.
- Evolution of cortical computation mechanisms in cognitive processes across different species.